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The water relations of the root–soil interface

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ABSTRACT

The difference in hydrostatic pressure between the xylem of the leaf and the soil depends, for a given transpiration rate, on the series of hydraulic resistances encountered along this pathway. Many studies have shown that the sum of the resistances in the plant and the soil is too small to account for the fall in water pressure between the leaf xylem and the soil, especially when plants are growing in sandy soils, which are prone to dry rapidly. A resistance at the root–soil interface, caused possibly by poor contact between the roots and the soil, has been proposed to account for the discrepancy. We explored the resistance in the pathway from soil to leaf using a technique that allows precise and continuous non-destructive measurement of the hydrostatic pressure in the leaf xylem. When the soil was leached with water, the fall in leaf water status as the soil dried was reasonably well described by a simple physical model without the need to invoke an interfacial resistance. However, when the soil was flushed with a nutrient solution with an osmotic pressure of 70 kPa, the hydrostatic pressure in the leaf xylem fell several times faster than that in the soil. We suggest that solutes accumulated either in the root or just outside it, creating large osmotic pressures, which gave the appearance of an interfacial resistance.

Key-words: hydraulic resistance; hydrostatic pressure; osmotic pressure; root water uptake; water transport.

INTRODUCTION

When water moves through the soil to the roots of a transpiring plant it usually does so by flowing viscously down gradients in hydrostatic pressure. If the soil is especially dry (with a matric suction greater than about 1·5 MPa) there may be a significant movement of water as vapour, but by then the transpiration rate is probably very low. Gradients in osmotic pressure move little water, because the transport coefficients for diffusion are typically orders of magnitude smaller than for viscous flow.

Movement across the interface between root and soil is more complicated. There may be a mucilaginous layer containing pores so small that the flow of water across it is greatly hindered, or poor hydraulic continuity between root and soil if the root is growing in a pore wider than itself or if the root has shrunk. Within the root itself water possibly flows viscously within the cell walls, but must then enter the symplast, at least temporarily, before leaving the cortex for the vascular cylinder. The passage across the semipermeable membranes bounding the symplast could lead to large effects on the osmotic pressure in the cortical cell walls if there is much exclusion of solutes at the membranes (Nulsen & Thurtell 1980).

These various processes and their implications for the uptake of water by roots have been discussed in a profusion of papers since the classical analysis by Gardner (1960) of the radial flow of water to a root. Despite the many useful contributions, a major conundrum remains. Sometimes the flow through root and soil can be neatly described in terms of viscous flow in the soil coupled with a simple hydraulic resistance in the root (e.g. Reicosky & Ritchie 1976; Blizzard & Boyer 1980; Passioura 1980). More often, a large interfacial resistance or other complication seems to be present (Herkelrath, Miller & Gardner 1977; Faiz & Weatherley 1978, 1982; Zur *et al.* 1982; Bristow, Campbell & Calissendorf 1984). This complication occurs especially with roots growing in sandy soils, which are prone to dry rapidly. Sudden drying could induce a self-amplifying shrinkage of the cortex, which would precipitate a loss of hydraulic continuity between root and soil (Passioura 1988), thereby creating a vapour gap (Huck, Klepper & Taylor 1970).

We have been exploring the water relations of the interface using a technique in which a pot is placed inside a pressure chamber, with the leaves of the plant outside the pressure chamber but enclosed in a cuvette so that their transpiration can be measured (Passioura 1980; Passioura & Munns 1984). When a pneumatic pressure is applied in the chamber, the large negative pressure that normally occurs in the xylem of the shoot is correspondingly raised. One can apply just enough pressure (the 'balancing pressure') to bring a cut in the leaf xylem to the point of bleeding, when the pressure in the xylem equals that of the atmosphere. The balancing pressure gives a precise and non-destructive measure of the fall in pressure between soil and leaf xylem, and in combination with the plant's transpiration rate enables the hydraulic properties of the system to be closely examined in a given plant as it dries the soil in which it is growing.

Our interest in the interface between the root and soil was kindled by a chance observation that well-fertilized plants had much more difficulty in extracting water from

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the soil than did moderately fertilized ones. Yet the difference in osmotic pressure of the soil solution between the two circumstances was but a small part of the normal drop in pressure across the plant. An interfacial effect seemed to be implicated.

MATERIALS AND METHODS

The soil

A sandy loam soil was collected from the field and packed into cylindrical PVC pots, 86 mm in diameter and 200 mm long, to a bulk density of 1·35 to 1·45 Mg m^{-3}. The particle size distribution of the soil comprised, on average, 55% coarse sand, 19% fine sand, 12% silt and 14% clay. The clay was non-swelling and largely kaolin. The function relating the water content to the suction in the soil was determined by using a pressure plate, and is shown (for drying soil) in Fig. 1. The diffusivity of soil water (D, m^2 s^{-1}) was determined as a function of water content by the technique of Rose (1968), and is also illustrated in Fig. 1.

The plants

A metal cap was fitted to the top of each pot, and a germinated tomato seed (*Lycopersicon esculentum* Mill. cv. UC82B) was sown through a small central hole in the cap. The hole in the cap around the stem was filled with silicone rubber after the plants were well established, to form a gas-tight seal. The plants were kept well watered and fertilized in a glasshouse for about 8 weeks with day/night temperatures of approximately 25/20 °C.

The plants were prepared for hydraulic measurement when their leaf areas were between 0·025 and 0·035 m^2. At this leaf area, a plant could dry the pot from about 20 kPa soil water suction to more than 300 kPa suction within 1 d. Before measurements of water relations were made, the pots were leached with 1000 cm^3 (twice the pore volume) of tap water, which had an osmotic pressure of 4 kPa, or with nutrient solution, which had an osmotic pressure of 70 kPa. The nutrient solution was a modified Hoagland's solution and contained, in mol m^{-3}, 6·5 K^+, 4·0 Ca^{2+}, 2·0 NH_4^+, 2·0 Mg^{2+}, 15·0 NO_3^-, 1·0 $H_2PO_4^-$, 2·0 SO_4^{2-}; and in mmol m^{-3}, 36·0 Fe^{3+}, 4·6 BO_4^{3-}, 0·5 Mn^{2+}, 0·2 Cu^{2+}, 0·2 Zn^{2+} and 0·1 $Mo_7O_{24}^{6-}$. During and after leaching the pots were placed on towels to facilitate drainage. In some experiments barley (*Hordeum vulgare* L. cv. Yagan) was used instead of tomato.

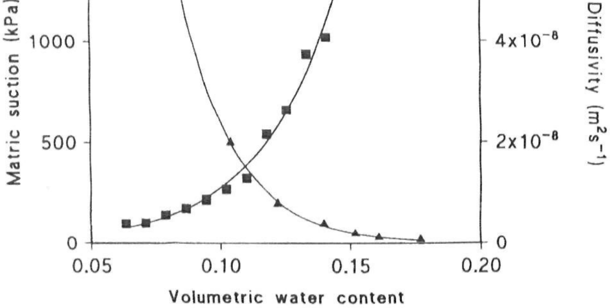

Figure 1. Matric suction (▲) and diffusivity (■) as functions of volumetric water content.

Measurements of water relations

The relation between transpiration rate and balancing pressure was determined essentially as described by Passioura & Munns (1984). In brief, a pot was put in a pressure chamber and the shoot of the plant, which remained outside the pressure chamber was enclosed in a water-jacketed cuvette. The plant was illuminated horizontally by a 400 W metal halide lamp which, with various amounts of shading with mesh screens, provided a photosynthetic photon flux density ranging from 0·4 to 2·0 mmol m^{-2} s^{-1}. Air was passed through the chamber, at rates ranging from 50 to 200×10^{-6} m^3 s^{-1}. The humidity of the air was measured with a dew-point hygrometer, and transpiration rate was determined by multiplying the flow rate through the cuvette by the difference in humidity between the ingoing and outgoing air.

Precisely enough gas pressure was applied in the root chamber to bring the xylem sap to the point of bleeding from a given cut in a leaf petiolule. We will henceforth call this pressure the 'balancing pressure', and interpret it to be numerically equal to the suction that would have existed in the xylem had the roots not been pressurized. Plants were kept at balancing pressure as the soil dried by using an automatic electronic controller with a sensor that determined the position of a meniscus in a water-filled tube that was attached to a petiolule of the leaf. If the hydrostatic pressure in the xylem fell below atmospheric pressure, the level of the miniscus fell and the gas pressure in the root chamber was increased. This technique was similar to that described by Passioura & Tanner (1985), except that the meniscus sensor was modified to provide a proportional rather than an on/off signal, thereby giving much finer control. It provided a continuous and non-destructive estimate of the suction in the leaf xylem. The equipment is shown schematically in Fig. 2.

Each experiment started with a plant in wet soil at a low transpiration rate. The transpiration rate was increased by changing the irradiance, humidity and flow rate of air. Typically, four step increases in transpiration were made and the corresponding balancing pressures were recorded. The plants were then kept at the highest transpiration rate, and the balancing pressure and soil water content were continuously recorded, until the balancing pressure approached 1500 kPa, which was the limit for the equipment we used.

The curve relating matric suction to water content in the soil was steep once the suction exceeded 200 kPa (Fig. 1). Calculations of matric suction from water content at the end of a drying cycle were prone to error because of slight differences between batches of soil. In order to test the

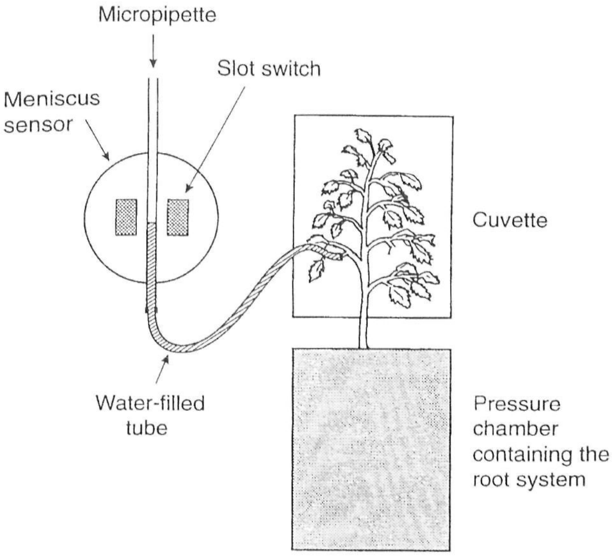

Figure 2. Schematic diagram of the experimental equipment. The slot switch consists of a light emitter and detector whose output depends on the level of the meniscus in the micropipette. An electronic controller adjusts the pressure in the root chamber to keep the meniscus stationary.

effect of different osmotic pressures of the soil solution on the ability of plants to get water out of the soil at equivalent matric suction, we carried out the measurements with high and low nutrient levels on the same plant under similar conditions. Thus, any error in the computation of matric suction, based on the weight of the pot, was common to both treatments. The experimental sequence was as follows: day 1, the pot was flushed with nutrient solution and drained on a towel overnight; day 2, the hydraulic behaviour of the soil–plant system was measured as the soil dried; day 3, the pot was leached with water and drained until the water content was the same as at the start of day 2; day 4, the hydraulic behaviour of the soil–plant system was measured under the same conditions as day 2. The order of leaching was reversed with some plants, i.e. leaching with water preceded that with nutrient solution.

Measurements at harvest

Each plant was harvested at the end of day 4 of the experiment, after hydraulic measurements had been made with the soil at high and low osmotic pressure. The soil water content was measured gravimetrically at three depths in the pot. Since the transpiration rate was measured continuously while hydraulic measurements were taken, and water content was fairly uniform throughout the pot, the average water content was back-calculated from this final value for each 10 min period during the experiment. Leaf area was measured (area measurement conveyor belt unit, Delta-T devices, Cambridge, UK) and roots were washed from the soil through a 2 mm sieve and their length measured (Comair root length scanner, Commonwealth Aircraft Corporation, Melbourne, Australia).

The osmotic pressure was measured from a bulked sample from the three depths. A saturated paste (30% water content) was prepared and left for 16 h before extraction of the soil solution by vacuum filtration. The osmolarity was measured using a micro-osmometer (3MO, Advanced Intruments, Needham Heights, MA, USA), and the osmotic pressure in the bulk soil solution was calculated based on the ratio of water contents in the soil and in the paste, on the assumption that the solutes did not react with the solid phase as the soil dried. We estimated the osmotic pressure following the first part of each sequence, i.e. day 2, by using the osmotic pressure of other pots containing plants of similar size, 48 h after they had been leached with water or nutrient solution.

THEORY

The balancing pressure (P_0) gives a precise and continuous estimate of hydrostatic pressure or suction in the leaf xylem. The suction in the leaf xylem is the sum of the bulk soil suction, τ, the mean pressure drop in the soil between the bulk and the root surface, the pressure drop in the plant between the root surface and the leaf xylem, plus any interfacial or osmotic effects.

$$P_0 = \tau_{\mathrm{soil}} + \Delta P_{\mathrm{soil}} + \Delta P_{\mathrm{plant}} + \text{interfacial and osmotic effects.} \qquad (1)$$

Since P_0 and τ_{soil} were measured and $\Delta P_{\mathrm{soil}} + \Delta P_{\mathrm{plant}}$ could be calculated (see below), any residual effect, which we assumed to be at the root–soil interface, could be quantified. In order to explore how the interfacial resistance changed as the soil dried, we plotted the suction at the surface of the root against the bulk matric suction in the soil. The suction at the root surface was estimated to be P_0–$\Delta P_{\mathrm{plant}}$, i.e the suction in the leaf xylem minus the calculated pressure drop through the plant.

Calculation of ΔP_{soil}

When the soil is wet, there is little resistance to the flow of water within it. However, substantial gradients in water content may develop in the soil near roots when the soil dries and the diffusivity (Fig. 1) becomes small. We calculated the difference in water content (or suction) between the surface of a root and the bulk soil by assuming cylindrical geometry and numerically solving the non-linear cylindrical diffusion equation in the manner described by Passioura & Cowan (1968), but modified to incorporate the powerful Jacobian analysis described by Campbell (1985). The equation and the appropriate initial and boundary conditions are:

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(rD(\theta)\frac{\mathrm{d}\theta}{\mathrm{d}r}\right), \qquad (2)$$

$$a \le r < b,\ \theta = \theta_0,\ t = 0,$$

$$r = a,\ D\frac{\mathrm{d}\theta}{\mathrm{d}r} = F,\ t > 0,$$

$$r = b,\ \frac{\mathrm{d}\theta}{\mathrm{d}r} = 0,\ t > 0,$$

where θ is the volumetric soil water content, θ_0 is the initial uniform θ, t is time, r is radial distance from the centre of a root, $D(\theta)$ is the diffusivity of soil water, a is the radius of the root and b is the radius of a cylinder of soil to which the average root is assumed to have exclusive access; $b=(\pi L)^{-1/2}$, where L is the length of root per volume of soil. F is the flux density ($m^3\ m^{-2}\ s^{-1}$) defined as the uptake rate of water ($m^3\ s^{-1}$) divided by the surface area of roots (m^2).

The equation can be solved to give θ at the surface of the root, which can be converted into a suction by means of the water release curve shown in Fig. 1. The pressure drop through the soil, ΔP_{soil}, is calculated as the difference between the suction at the root surface and the matric suction in the bulk soil.

Calculation of ΔP_{plant}

When the soil was wet, the balancing pressure was measured over a range of transpiration rates. As mentioned earlier, we assume that the balancing pressure equates to the suction that would have existed in the xylem had the roots not been pressurized. Figure 3 shows an example of the relation between this estimated xylem suction, P_o (kPa), and transpiration rate, E (mg s^{-1}), for a tomato plant. The slope of the regression line represents the hydraulic resistance to the flow of water through the soil and plant. The relation is typically linear, and so the hydraulic behaviour of the plants can be neatly summarized in terms of the slope and intercept of the line. The intercept on the pressure axis is most simply interpreted as the soil water suction experienced by the roots (Passioura 1980; Passioura & Munns 1984). By assuming that the hydraulic resistance of the plant remained constant during an experiment, we could calculate ΔP_{plant} as the product of the transpiration rate and the resistance.

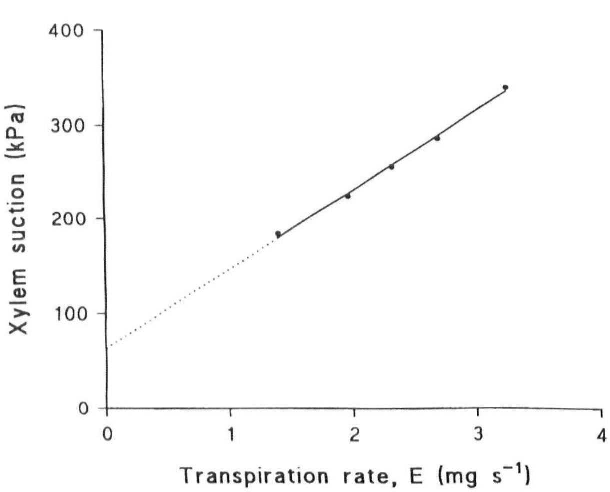

Figure 3. Suction in the xylem of a tomato plant plotted as a function of increasing transpiration rate.

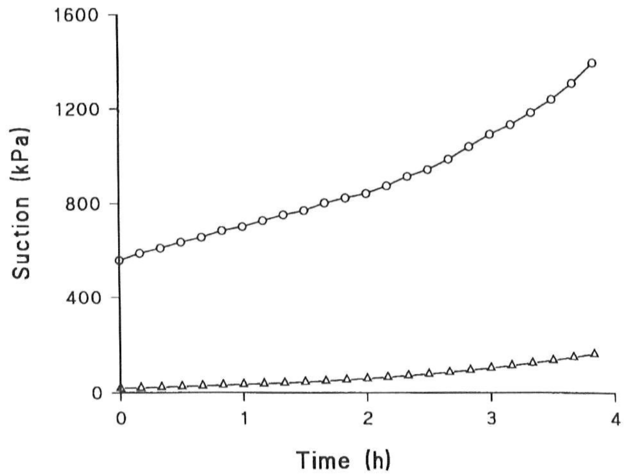

Figure 4. The suction in the leaf xylem (○) and in the soil (△) as functions of time.

RESULTS AND DISCUSSION

Figure 4 shows the suction in the leaf xylem as a function of time for a rapidly transpiring plant when the soil had been flushed with nutrient solution. The evident rise in xylem suction is to be expected for two reasons. First, the continued drying of the soil increases the bulk soil water suction. Second, the drying also decreases the diffusivity (Fig. 1), thereby increasing the local gradients in suction around the roots, which in wet soil are trivial, but which can be large in dry soil. During the 4 h for which the plant was transpiring, the balancing pressure rose by 780 kPa, yet the bulk matric suction rose by only 140 kPa.

Figure 5 describes a similar experiment to that in Fig. 4, but the axes have been changed to show the relationship between suction at the root surface and suction in the bulk soil. The x-axis now represents bulk matric suction in the soil during the drying period rather than time. The suction at the root surface has been calculated from the balancing pressure (P_0) as $P_0 - \Delta P_{plant}$, as described by Eqn 1. Figure 5 shows the root suction as a function of bulk matric suction for the same plant transpiring under similar conditions on two different days. In the first case the soil had been flushed with nutrient solution, and in the second the soil had been leached with water.

Figure 5 also shows the suction at the root surface calculated from numerical solutions of the cylindrical diffusion equation (Eqn 2). The numerical analysis required three inputs: the transpiration rate (which sometimes fell substantially as the soil dried, and which was measured), the root radius (which averaged 250 μm) and the root length density. We measured the total root length at the end of the experiment, but it is likely that only part of the root system was taking up water, so that the effective root length was perhaps much smaller than the measured length. Hence we solved the equation for different values of root length density, namely total (40 km m^{-3}), one-half, one-quarter and one-tenth of the total. The calculated root suction for each root length is shown as the dashed lines in Fig. 5. The

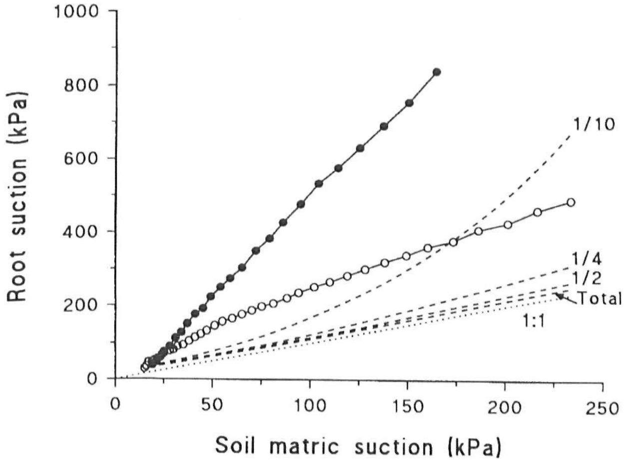

Figure 5. The root suction of tomato plants in soil flushed with nutrient solution (●) or leached with water (○), calculated from experimental measurements. The dashed lines show the calculated root suction based on Eqn 2 for the total, one-half, one-quarter and one-tenth of the measured root length.

dotted line on Fig. 5 represents a 1:1 relationship between the suction at the root surface and the matric suction of the bulk soil. The deviations from the dotted (1:1) line and the calculated root suctions for the four different root lengths reflect the increasing soil resistance, as the fall in diffusivity gives rise to local gradients in suction around the roots.

There was no agreement between the root suction calculated from experimental measurements and the calculations based on Eqn 2 for the nutrient-flushed soil when we assume that more than one quarter of the root length was functioning. Even if we assume that only one tenth of the roots were taking up water, the calculated values of suction at the root surface were far too low. This discrepancy cannot be accounted for by an increase in the bulk osmotic pressure of the soil solution. At the end of the experiment the osmotic pressure of the soil solution was 120 kPa. If no nutrients were removed during the 4 h duration of the experiment, we calculate that the osmotic pressure at the start would have been 80 kPa. Thus the maximum increase in osmotic pressure would not have exceeded 40 kPa.

For the water-leached soil, there was a closer agreement at higher root lengths between root suction calculated from experimental measurements and the calculations based on Eqn 2, apart from a puzzling initial rise in root suction when the soil was still wet. The proposition that less than one tenth of the root length was taking up water, which seemed faintly plausible in the case of the nutrient-flushed soil, does not hold for water-leached soil because the local gradients in the soil around roots would be too great once the soil suction increased above 200 kPa. This point is further illustrated in Fig. 6, which is a continuation of the data for the water-leached soil of Fig. 5. The plant in water-leached soil was able to dry the soil to a larger matric suction before the pressure limit of the equipment was reached, compared with the plant in nutrient-flushed soil. The calculations suggest that by the time the soil matric suction reached 500 kPa, over one-quarter of the measured root length would have still been taking up water.

Given that Fig. 5 depicts the behaviour of the same plant transpiring at similar rates, the effect of the seemingly subtle difference in the composition of the soil solution is startling. The experiment above was repeated with the soil leached with water first and nutrient solution second (the reverse of the above). In this case the osmotic pressure of the nutrient-flushed soil increased from about 40 to 70 kPa as the soil dried. This was lower than the nutrient–water experiment shown in Fig. 5, probably as a consequence of the demand for nutrients, which was so high after the pot had been leached with water that the roots rapidly depleted the soil solution of nutrients when these were eventually supplied. The root suction as a function of soil matric suction increased twice as fast when the soil was flushed with nutrient solution compared with when it was flushed with water. This was not as spectacular as that in Fig. 5, when the osmotic pressure was about 50 kPa higher, but still it remains striking that such weak soil solutions necessitate such large suctions to maintain transpiration.

Figure 7 shows root suction in relation to soil matric suction for a barley plant. The barley plants had a lower leaf area than the tomatoes, and consequently it took several days for the barley to extract a similar amount of water from the soil. At the start of the experiment the osmotic pressure of the soil solution was about 120 kPa. Days 1–5 show the relationship between soil matric suction and balancing pressure when the plants were kept at a high transpiration rate during the day. The lack of spread of values on days 1 and 3 results from the shape of the water release curve (Fig. 1), which has a small slope in wet soil. At the end of day 5 the soil was leached with water. The plant was then allowed to dry the soil to 160 kPa suction over the next 7 d before the final run was made at an osmotic pressure of 30 kPa. Again we see the dramatic effect the osmotic pressure of the soil solution had on the apparent hydraulic resistance of the system. In fact, the

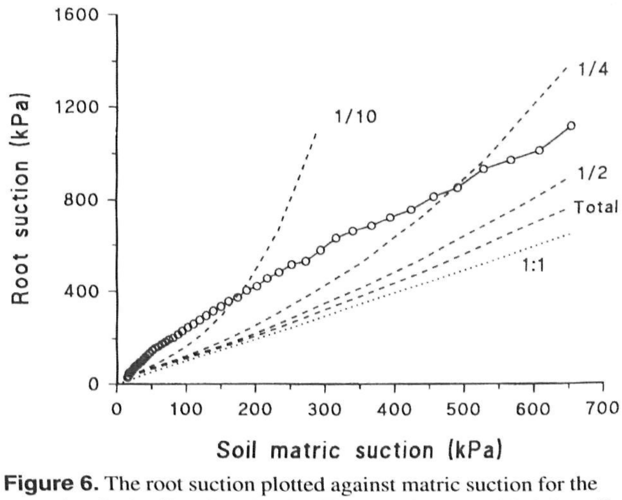

Figure 6. The root suction plotted against matric suction for the water-leached soil as described in Fig. 5, as the plant dried the soil further to 650 kPa suction.

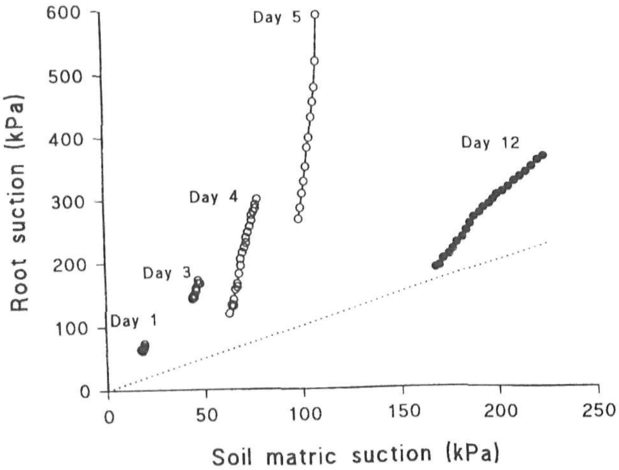

Figure 7. The root suction of a barley plant calculated from experimental measurements (○) plotted against soil matric suction over a 5 d drying cycle, and again on day 12 after the soil was leached with water and permitted to dry for a further 7 d (●).

balancing pressure of the plant in nutrient-flushed soil increased eight times faster than that in water-leached soil, as the soil dried.

GENERAL DISCUSSION

Leaching the soil with water had a very large effect on the ability of the plants to extract water. Plants in soil leached with water dried the soil to the point where the diffusivity was low and substantial localized gradients in water content probably developed around individual roots (Fig. 6). In contrast, the plants in nutrient-flushed soil had difficulty in extracting water from the soil and large differences in suction developed between the leaf and soil. Three possibilities for this different behaviour come to mind; first, that the hydraulic resistance of the plants in nutrient-flushed soil greatly increased over time; second, that a large interfacial resistance developed between roots and soil because of poor root–soil contact; and third, that the presence of the extra solutes resulted in a large rise in the osmotic pressure at the surface of the roots or possibly within the apoplast of the roots.

The first possibility, namely that the hydraulic resistance of the plant was affected by the presence of solutes, is unlikely. Our results show no relationship between hydraulic resistance and osmotic pressure in the soil when the soil was wet (data not shown), in agreement with Shalhevet *et al.* (1976) and Munns & Passioura (1984). The second argument, that of poor root-soil contact or an interfacial resistance, has been used to account for the discrepancy between the total hydraulic resistance between the soil and leaf of a transpiring plant and the sum of the individual plant and soil resistances (Herkelrath *et al.* 1977; Faiz & Weatherley 1978, 1982; Bristow *et al.* 1984; Jensen, Henson & Turner 1989). In our experiments we would need to invoke a very large interfacial resistance in the case of the nutrient-flushed plant and only a small interfacial resistance in the case of a water-leached plant. It is difficult to see how differences in the osmotic pressure of the soil solution surrounding the roots would affect root–soil contact.

It is likely therefore that the 'missing' resistance in the fertilized plants reflected a rise in osmotic pressure at the interface between the root and the soil, or perhaps in the apoplast of the root. A large build-up of solutes at the interface between root and soil or within the cortex of the root has been calculated or observed (Passioura & Frere 1967; Riley & Barber 1970; Sinha & Singh 1974; Nulsen & Thurtell 1980; Schleiff 1983; Hamza & Aylmore 1992), but generally under conditions of much higher soil osmotic pressures than were experienced by the tomatoes in our experiments. Our nutrient solution had an osmotic pressure of 70 kPa and the soil solution presumably had this osmotic pressure at the start of an experiment. The greatest difference in osmotic pressure between the water-leached and nutrient-flushed soil as the soil dried in the tomato experiments was less than 100 kPa. How could such small differences between water and nutrient treatments result in the large discrepancies evident in Fig. 5?

We postulate two hypotheses, summarized in Fig. 8, to explain a local build-up in osmotic pressure, namely a build-up outside the root or a build-up inside the root cortex.

(1) An accumulation of solutes at the surface of the roots (Riley & Barber 1970; Sinha & Singh 1974). This would be expected when the root epidermis excludes some of the solutes carried towards it in the transpiration stream. Large amplifications of the small differences in osmotic pressure of the soil solution between the treatments could arise if the diffusion coefficients, D_s, for the solutes in the soil were very small. Few data for such diffusion coefficients are available, but if D_s values were of the order of 3×10^{-11} m^2 s^{-1}, which is within the range reported by So & Nye (1989), then a large amplification is plausible (see Fig. 3 in Passioura & Frere 1967). Such a build-up remains unlikely in wet soil, in which D_s is much larger.

(2) An accumulation of solutes in the apoplast of the root (Nulsen & Thurtell 1980; Canny & Huang 1993). This might occur if the hydraulic continuity between root and soil was large enough to allow substantial influx of nutrients by mass flow in the nutrient stream, but small enough to inhibit the diffusion of solutes out of the apoplast back into the soil across point contacts between soil particles and the root surface (Passioura 1988).

These arguments are plausible when a large proportion of the incoming solutes are excluded by the root, for example when the concentration of nutrients is high or sodium chloride is the main solute. They are less plausible in the case of the tomato plant in our experiments which had been leached with water before being flushed with nutrient solution, for the roots took up much of the nutrient supplied in 2 d. But recent work by Canny and associates

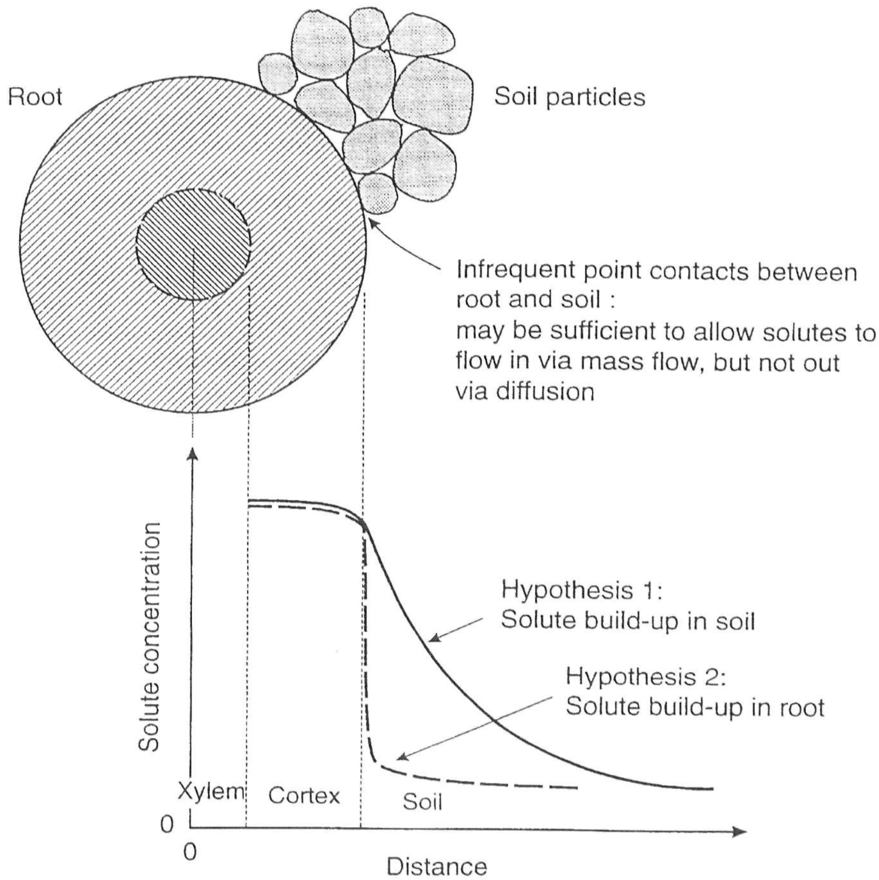

Figure 8. Possible locations for the build up of solutes around a root.

(eg. Varney & Canny 1993) has shown that nutrients and water may be taken up by different parts of the root system. Thus, although the root system *as a whole* is taking up nutrients, there may still be a substantial build-up at those parts of the interface between root and soil where there is a putative uptake of water but not of nutrients.

Our experiments suggest that the seemingly large interfacial resistances that have been reported may in fact have arisen because of local osmotic rather than hydraulic effects, despite the low bulk osmotic pressures of the soil solution. If so, it is inappropriate to model the uptake of water as though it were strongly influenced by an interfacial resistance (De Willigen & Van Noordwijk 1987; Jensen *et al.* 1993). Some of the controversy surrounding the nature of hydraulic resistance may be resolved by looking at the time period over which it was measured. Rapid measurements would tend to give a constant resistance, but over longer periods there would be time for osmotic gradients to build up.

The osmotic pressures used in our experiments, 40–120 kPa, are not unusually large and are common in agricultural soils (Russell 1988). Thus, the phenomenon that we have described, that plants may have greater difficulty in extracting water from soil than would be expected from summing bulk osmotic and matric potentials, may apply in the field as well as in the laboratory.

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